

FORENSIC INVESTIGATION REPORT

CyberDefenders — Lockdown Lab

Lab Platform	CyberDefenders.org
Category	Network Forensics / DFIR
Difficulty	Easy
Artifacts	PCAP, Memory Dump (memdump.mem), Malware Sample (updatenow.exe)
Tools Used	Wireshark, Volatility 3, FLOSS/Strings, VirusTotal, MalwareBazaar
MITRE ATT&CK Tactics	Execution, Persistence, Defense Evasion, Discovery, Lateral Movement, C2

1. Executive Summary

TechNova Systems' SOC detected suspicious outbound traffic from a public-facing IIS server. This investigation reconstructs a multi-stage intrusion beginning with network reconnaissance and culminating in the deployment of an AgentTesla Remote Access Trojan (RAT). The attacker (10.0.2.4) performed port scanning, accessed SMB shares to upload a web shell, established a reverse shell, planted a persistence implant, and installed a packed, commodity RAT that beacons to a Ukrainian C2 domain.

Key Findings at a Glance:

- Attacker IP: 10.0.2.4 — conducted port/service reconnaissance against 10.0.2.15
- Web shell (shell.aspx, 1,015,024 bytes) uploaded via SMB to the IIS Startup directory
- Reverse shell established on port 4443 (firewall-friendly / non-standard)
- Persistence implant: updatenow.exe placed in Windows Startup folder (T1547)
- Malware packed with UPX; identified as AgentTesla RAT beacons to cp8nl.hyperhost.ua

2. Phase 1 — PCAP Analysis

Three questions were answered using Wireshark against the provided capture.pcap. The analysis covers reconnaissance, SMB lateral movement, web shell delivery, and reverse shell callback.

Q1 — Reconnaissance Origin

Q1 After flooding the IIS host with rapid-fire probes, the attacker reveals their origin. Which IP address generated this reconnaissance traffic?

Answer 10.0.2.4

Method Wireshark → Statistics → Conversations. IP 10.0.2.4 generated an anomalous volume of SYN packets to sequential destination ports on 10.0.2.15, confirming port-scan reconnaissance.

The screenshot shows the Wireshark interface with the 'Statistics - Conversations' window open. The 'Conversations' pane on the left shows a list of conversations. The 'Packets' column for the conversation between 10.0.2.4 and 10.0.2.15 is highlighted with a red box, showing 6,007 packets. The 'Bytes' column for the same conversation shows 4 MB. The 'Stream ID' column shows 6. The 'Packets A → B' column shows 4,166 packets. The 'Bytes A → B' column shows 3 MB. The 'Packets B → A' column shows 1,841 packets. The 'Bytes B → A' column shows 184 kB. The 'Rel Start' column shows 78.082576. The 'Duration' column shows 1434.6036. The 'Bits/s A → B' column shows 19 kbps. The 'Bits/s B → A' column shows 1,028 bi.

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.0.2.4	10.0.2.15	6,007	4 MB	6	4,166	3 MB	1,841	184 kB	78.082576	1434.6036	19 kbps	1,028 bi
10.0.2.15	10.0.2.3	10	5 kB	7	5	2 kB	5	3 kB	84.517280	1199.3760	11 bits/s	19 bi
10.0.2.15	10.0.2.255	2	486 bytes	12	2	486 bytes	0	0 bytes	145.653875	720.2510	5 bits/s	0 bi
10.0.2.15	20.42.73.29	24	11 kB	13	11	5 kB	13	6 kB	573.288138	1.2100	36 kbps	37 k
10.0.2.15	20.198.118.190	5	550 bytes	14	2	207 bytes	3	343 bytes	608.774473	343.0272	4 bits/s	7 bi
10.0.2.15	20.198.119.143	21	8 kB	15	11	3 kB	10	5 kB	1030.943915	0.3470	66 kbps	123 k
10.0.2.15	20.242.39.171	35	20 kB	2	16	13 kB	19	7 kB	0.957982	3.4821	29 kbps	15 k
10.0.2.15	23.58.93.34	24	3 kB	4	14	2 kB	10	2 kB	53.750251	63.2284	228 bits/s	208 bi
10.0.2.15	40.81.94.65	14	1 kB	5	7	630 bytes	7	630 bytes	77.532248	1280.2031	3 bits/s	3 bi
10.0.2.15	52.252.198.177	4	228 bytes	0	2	108 bytes	2	120 bytes	0.000000	0.2820	3,063 bits/s	3,403 bi
10.0.2.15	104.26.10.240	4	228 bytes	11	2	108 bytes	2	120 bytes	102.344396	0.0098	87 kbps	97 k
10.0.2.15	192.168.1.1	52	5 kB	1	46	4 kB	6	1 kB	0.948048	1029.9916	28 bits/s	8 bi
10.0.2.15	216.58.200.163	12	2 kB	3	7	792 bytes	5	734 bytes	53.663407	63.3154	100 bits/s	92 bi
10.0.2.15	224.0.0.22	25	1 kB	8	25	1 kB	0	0 bytes	84.546591	1199.7954	9 bits/s	0 bi
10.0.2.15	224.0.0.251	14	1 kB	9	14	1 kB	0	0 bytes	84.560786	1199.3855	8 bits/s	0 bi
10.0.2.15	224.0.0.252	7	462 bytes	10	7	462 bytes	0	0 bytes	84.567243	1199.3847	3 bits/s	0 bi

Q2 — Reconnaissance Technique (MITRE ATT&CK)

Q2 Zeroing in on a single open service to gain a foothold, the attacker carries out targeted enumeration. Which MITRE ATT&CK technique ID covers this activity?

Answer T1046 — Network Service Discovery

Method The systematic probing of sequential ports maps directly to T1046 (Network Service Discovery) in the MITRE ATT&CK framework, used by adversaries to enumerate open services and identify attack vectors.

The image shows a Wireshark packet capture of a network traffic analysis. The top pane displays a list of captured packets, with a filter applied: `ip.src == 10.0.2.4 && tcp`. The packet list shows a series of TCP SYN packets from source 10.0.2.4 to destination server.local on port 113. The packet details pane on the left shows the structure of a selected packet (No. 74), including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol (TCP) fields. The TCP field shows a SYN flag set, sequence number 0, and window size 1024. The packet bytes pane on the right shows the raw data of the packet, including the TCP header and payload.

No.	Time	Source	Destination	Protocol	Length	Info
74	78	10.0.2.4	server.local	TCP	60	55475 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
76	78	10.0.2.4	server.local	TCP	60	55475 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
78	78	10.0.2.4	server.local	TCP	60	55475 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
79	78	10.0.2.4	server.local	TCP	60	55475 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
82	78	10.0.2.4	server.local	TCP	60	55475 → 135 [RST] Seq=1 Win=0 Len=0
83	78	10.0.2.4	server.local	TCP	60	55475 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
85	78	10.0.2.4	server.local	TCP	60	55475 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
86	78	10.0.2.4	server.local	TCP	60	55475 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
89	78	10.0.2.4	server.local	TCP	60	55475 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
91	78	10.0.2.4	server.local	TCP	60	55475 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
93	78	10.0.2.4	server.local	TCP	60	55475 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
94	78	10.0.2.4	server.local	TCP	60	55475 → 445 [RST] Seq=1 Win=0 Len=0
96	78	10.0.2.4	server.local	TCP	60	55475 → 1720 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
98	78	10.0.2.4	server.local	TCP	60	55475 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
100	78	10.0.2.4	server.local	TCP	60	55475 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
101	78	10.0.2.4	server.local	TCP	60	55475 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
104	78	10.0.2.4	server.local	TCP	60	55475 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
105	78	10.0.2.4	server.local	TCP	60	55475 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Frame 74: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface
Ethernet II, Src: PCSystemtec 6b:6e:c6 (08:00:27:6b:6e:c6), Dst: server.local (10.0.2.15)
Internet Protocol Version 4, Src: 10.0.2.4 (10.0.2.4), Dst: server.local (10.0.2.15)
Transmission Control Protocol, Src Port: 55475, Dst Port: 113, Seq: 0, Len: 0
Source Port: 55475
Destination Port: 113
[Stream index: 5]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (37)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3169723535
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
010 ... = Header Length: 24 bytes (6)
Flags: 0x002 (SYN)
Window: 1024
[Calculated window size: 1024]
Checksum: 0x174 (unverified)

Source Port (tcp.srcport), 2 bytes

Packets: 6359 · Displayed: 4160 (65.4%)

Profile: Default

Q3 — SMB Share Enumeration

Q3 While reviewing the SMB traffic, you observe two consecutive Tree Connect requests that expose the first shares the intruder probes. Which two full UNC paths are accessed?

Answer \\10.0.2.15\Documents and \\10.0.2.15\IPC\$

Method Filter: ip.addr == 10.0.2.4 && smb2. Statistics → Protocol Hierarchy confirmed SMB2. Two consecutive SMB2 Tree Connect Request packets revealed the UNC paths: \\10.0.2.15\Documents and \\10.0.2.15\IPC\$.

The screenshot shows a Wireshark capture of SMB traffic. The filter is set to 'ip.src == 10.0.2.4 && smb2'. The packet list shows a sequence of SMB2 messages. Packet 2678 is highlighted, showing a Tree Connect Request to '\\10.0.2.15\Documents'. The packet details pane shows the SMB2 header and the Tree Connect Request structure, including the path '\\10.0.2.15\Documents'.

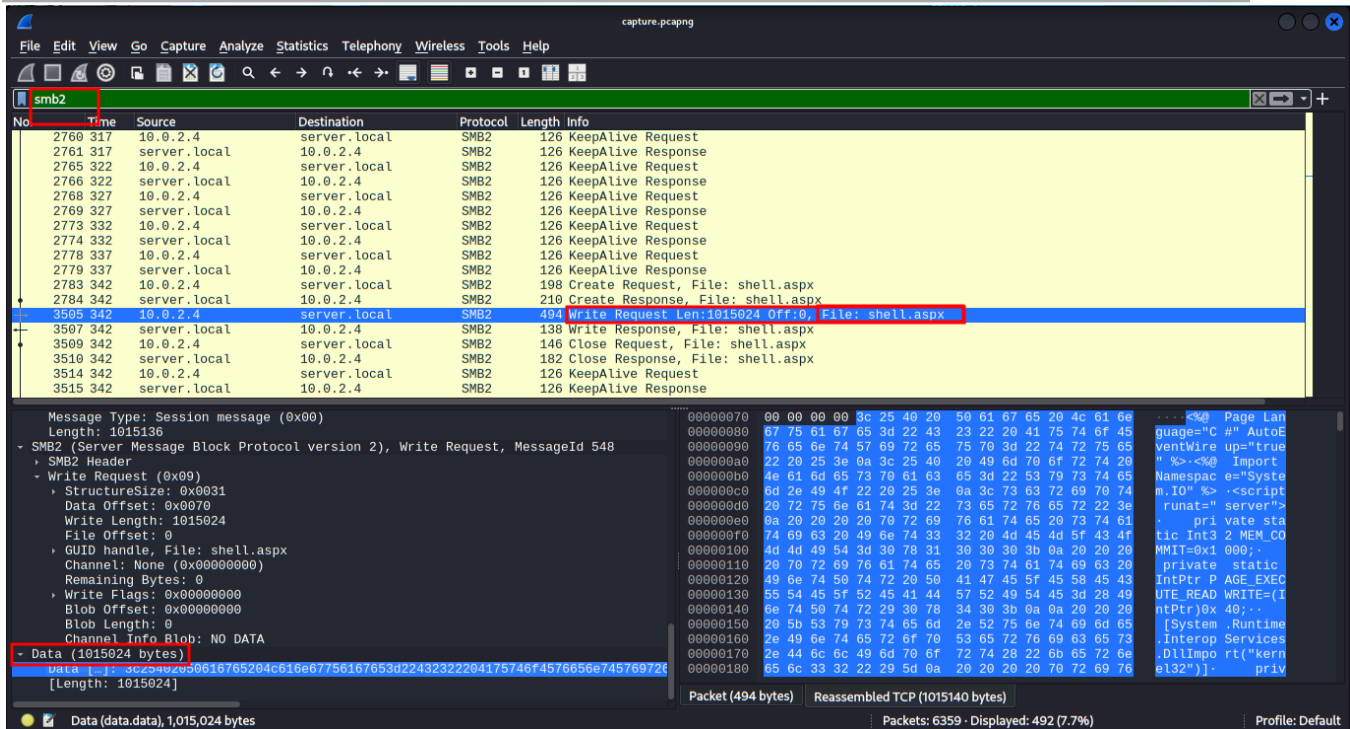
The screenshot shows a Wireshark capture of SMB traffic. The filter is set to 'smb2'. The packet list shows a sequence of SMB2 messages. Packet 2678 is highlighted, showing a Tree Connect Request to '\\10.0.2.15\IPC\$'. The packet details pane shows the SMB2 header and the Tree Connect Request structure, including the path '\\10.0.2.15\IPC\$'.

Q4 — Web Shell Upload

Q4 Inside the share, the attacker plants a web-accessible payload that will grant remote code execution. What is the filename and byte length specified in the SMB2 Write Request?

Answer	shell.aspx 1,015,024 bytes
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Method	Wireshark → File → Export Objects → SMB revealed shell.aspx. The corresponding SMB2 Write Request packet confirmed a data length of 1,015,024 bytes. The ASPX web shell provides remote code execution on the IIS host.
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Q5 — Reverse Shell Port

Q5 The newly planted shell calls back to the attacker over an uncommon but firewall-friendly port. Which listening port did the attacker use for the reverse shell?

Answer 4443

Method Filtered traffic for shell.aspx web requests, then followed subsequent packets. Packet 3573 and surrounding traffic showed an outbound TCP connection from the IIS server (10.0.2.15) back to attacker (10.0.2.4) on destination port 4443.

The image shows a Wireshark packet capture of a network traffic. The top bar indicates the capture file is 'capture.pcapng'. The filter bar shows 'ip.src == 10.0.2.4 && tcp'. The packet list shows a series of packets from 10.0.2.4 to server.local. Packet 3573 is highlighted, showing a TCP connection from 10.0.2.4 to 10.0.2.15 on port 4443. The packet details pane shows the following information:

- Identification: 0x0000 (0)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: TCP (6)
- Header Checksum: 0x22b2 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.0.2.4 (10.0.2.4)
- Destination Address: server.local (10.0.2.15)
- [Stream index: 6]
- Transmission Control Protocol, Src Port: 4443, Dst Port: 49688, Seq: 0, Ack: 1, Len: 0
- Source Port: 4443
- Destination Port: 49688
- [Stream index: 1081]
- [Stream Packet Number: 2]
- [Conversation completeness: Incomplete, DATA (15)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 686761803
- [Next Sequence Number: 1 (relative sequence number)]

The packet bytes pane shows the raw data of the packet, including the TCP header and the data payload.

3. Phase 2 — Memory Dump Analysis

The memory image (memdump.mem) was analyzed using Volatility 3. Three questions required kernel introspection, process enumeration, and network connection correlation.

Q6 — Kernel Base Address

Q6 Your memory snapshot captures the system's kernel in situ. What is the kernel base address in the dump?	
Answer	0xf80079213000
Method	Command: vol.py -f memdump.mem windows.info — The KernBase field in the Volatility output directly reports the kernel base address, providing critical context for validating the image profile and KASLR offset.

```
(kn999@kn999) ~[~/volatility3]
$ python3 vol.py -f /home/kn999/Downloads/269-lockdown/memdump.mem windows.info
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0xf80079213000
Dir 0x1a0000
Symbols file:///home/kn999/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/EF9A48AFA50FF07C6165858B01919536-1.json.xz
Is64Bit True
IsPAE False
Layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf80079613f10
Major/Minor 15.17763
MachineType 34404
KeNumberProcessors 4
SystemTime 2024-09-10 06:14:13+00:00
NtSystemRoot C:\Windows
NtProductType NtProductServer
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sun Nov 10 07:20:39 2075
```

Q7 — Persistence Implant Path & ATT&CK ID

Q7 A trusted service launches an unfamiliar executable outside the usual IIS stack. What is the full on-disk path of that executable and which MITRE ATT&CK persistence technique ID corresponds?

Answer C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe | T1547

Method Commands: vol.py -f memdump.mem windows.cmdline and windows.psTree. The w3wp.exe process (PID 4332) spawned updatenow.exe from the Windows Startup folder — a classic T1547.001 (Boot/Logon Autostart Execution: Registry Run Keys / Startup Folder) persistence technique.

```
kn999@kn999: ~/volatility3
Session Actions Edit View Help
kn999@kn999: ~/volatility3
** 2304 628 svchost.exe 0xc0654d4e0c0 16 - 0 False 2024-09-10 05:30:04.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvc -p -s Winmgmt C:\Windows\system32\svchost.exe
** 2180 628 svchost.exe 0xc0656dd080 3 - 1 False 2024-09-10 05:31:50.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k UnistackSvcGroup -s CDPUserSvc C:\Windows\system32\svchost.exe
** 776 628 svchost.exe 0xc0657050800 3 - 0 False 2024-09-10 05:29:50.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k NetworkServiceNetworkRestricted -p -s PolicyAgent C:\Windows\system32\svchost.exe
** 2440 628 svchost.exe 0xc06571a3300 4 - 0 False 2024-09-10 05:30:04.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalService -s W32Time C:\Windows\system32\svchost.exe
** 908 628 svchost.exe 0xc0654fa4080 13 - 0 False 2024-09-10 05:29:14.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k RPCSS -p C:\Windows\system32\svchost.exe
** 1292 628 svchost.exe 0xc0656de0800 3 - 0 False 2024-09-10 05:29:27.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvc -p -s gsvc C:\Windows\system32\svchost.exe
** 780 628 svchost.exe 0xc0657052080 5 - 0 False 2024-09-10 05:29:50.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvc -p -s IKEEXT C:\Windows\system32\svchost.exe
** 1680 628 svchost.exe 0xc0656eb4080 4 - 0 False 2024-09-10 05:29:34.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s FontCache C:\Windows\system32\svchost.exe
** 1808 628 svchost.exe 0xc0656f94080 5 - 0 False 2024-09-10 05:29:38.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k NetworkService -p -s LanmanWorkstation C:\Windows\system32\svchost.exe
** 3856 628 svchost.exe 0xc0657309080 7 - 0 False 2024-09-10 05:32:27.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s CDPSvc C:\Windows\system32\svchost.exe
** 2064 628 svchost.exe 0xc06578ed080 6 - 0 False 2024-09-10 05:44:23.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s StorSvc C:\Windows\system32\svchost.exe
** 660 628 svchost.exe 0xc06566c30c0 5 - 0 False 2024-09-10 05:29:22.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s NcbService C:\Windows\system32\svchost.exe
** 2452 628 svchost.exe 0xc06571cb280 15 - 0 False 2024-09-10 05:30:04.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k iisvc C:\Windows\system32\svchost.exe
*** 4332 2452 w3wp.exe 0xc06574ca080 0 - 0 False 2024-09-10 05:44:45.000000 UTC 2024-09-10 06:10:48.000000 UTC \Device\HarddiskVolume1\Windows\System
32\inetstrv\w3wp.exe
*** 900 4332 updatenow.exe 0xc0657dd080 3 - 0 True 2024-09-10 05:33:42.000000 UTC N/A \Device\HarddiskVolume1\ProgramData\Microsoft\Windows\Start Me
nu\Programs\Startup\updatenow.exe "C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe"
** 792 628 svchost.exe 0xc0654f69080 14 - 0 False 2024-09-10 05:29:12.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k DcomLaunch -p C:\Windows\system32\svchost.exe
** 2696 792 SearchUI.exe 0xc065730d080 14 - 1 False 2024-09-10 05:33:42.000000 UTC N/A \Device\HarddiskVolume1\Windows\SystemApps\Microsoft.Windows.C
ortana_cw5n1h2txyewy\SearchUI.exe "C:\Windows\SystemApps\Microsoft.Windows.Cortana_cw5n1h2txyewy\SearchUI.exe" -ServerName:CortanaUI.AppXa50dqa5gqv4428c9y1jw7m3btvej.mca
C:\Win
dows\SystemApps\Microsoft.Windows.Cortana_cw5n1h2txyewy\SearchUI.exe
** 1704 792 RuntimeBroker.exe 0xc06579ad080 1 - 1 False 2024-09-10 05:34:01.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\RuntimeBroker.exe C
:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
** 4092 792 ShellExperienceHost.exe 0xc06572b080 13 - 1 False 2024-09-10 05:33:35.000000 UTC N/A \Device\HarddiskVolume1\Windows\SystemApps\ShellExperienceHost
_cw5n1h2txyewy\ShellExperienceHost.exe "C:\Windows\SystemApps\ShellExperienceHost_cw5n1h2txyewy\ShellExperienceHost.exe" -ServerName:AppXtk181tbx0c2q5x02s8tw7hfxa9xb3t.mca
C:\Win
dows\SystemApps\ShellExperienceHost_cw5n1h2txyewy\ShellExperienceHost.exe
** 2036 792 RuntimeBroker.exe 0xc0656dd40c0 3 - 1 False 2024-09-10 05:33:46.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\RuntimeBroker.exe C
:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
** 2072 628 svchost.exe 0xc065703e080 8 - 0 False 2024-09-10 05:29:52.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvc -p -s UserManager C:\Windows\system32\svchost.exe
** 2172 2072 sihost.exe 0xc06571d5080 5 - 1 False 2024-09-10 05:31:50.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\sihost.exe sihost
.exe C:\Windows\system32\sihost.exe
** 1308 628 svchost.exe 0xc0656ddc080 3 - 0 False 2024-09-10 05:29:27.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvc -p -s Themes C:\Windows\System32\svchost.exe
```

```
2304 svchost.exe C:\Windows\system32\svchost.exe -k netsvc -p -s Winmgmt
2340 svchost.exe C:\Windows\system32\svchost.exe -k LocalService -p -s SstpSvc
2360 svchost.exe C:\Windows\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s SysMain
2404 svchost.exe C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s TrkWks
2440 svchost.exe C:\Windows\system32\svchost.exe -k LocalService -s W32Time
2452 svchost.exe C:\Windows\system32\svchost.exe -k iisvc
2492 svchost.exe C:\Windows\system32\svchost.exe -k netsvc -p -s WpnService
2516 svchost.exe C:\Windows\System32\svchost.exe -k NetworkService -p -s WinRM
2552 wlm.exe C:\Windows\system32\wlm\wlm.exe
2632 MsMpEng.exe "C:\ProgramData\Microsoft\Windows Defender\platform\4.18.24070.5-0\MsMpEng.exe"
2760 svchost.exe C:\Windows\System32\svchost.exe -k smbvc -s LanmanServer
2784 svchost.exe C:\Windows\System32\svchost.exe -k NetSvc -p -s iphlpsvc
2868 svchost.exe C:\Windows\System32\svchost.exe -k netsvc
2180 svchost.exe C:\Windows\system32\svchost.exe -k UnistackSvcGroup -s CDPUserSvc
2172 sihost.exe sihost.exe
1640 svchost.exe C:\Windows\system32\svchost.exe -k UnistackSvcGroup -s WpnUserService
752 taskhostw.exe {222A245B-E637-4AE9-A93F-A59CA119A75E}
2472 MicrosoftEdgeU C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /c
1748 svchost.exe C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s TabletInputService
928 ctfmon.exe "ctfmon.exe"
3580 svchost.exe C:\Windows\system32\svchost.exe -k netsvc -p -s TokenBroker
3508 svchost.exe C:\Windows\system32\svchost.exe -k appmodel -p -s StateRepository
3856 svchost.exe C:\Windows\system32\svchost.exe -k LocalService -p -s CDPSvc
3908 userinit.exe -
3976 explorer.exe C:\Windows\Explorer.EXE
4028 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceNoNetwork -p -s DPS
2244 msdtc.exe C:\Windows\System32\msdtc.exe
1892 svchost.exe C:\Windows\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s UALSVC
412 svchost.exe C:\Windows\system32\svchost.exe -k netsvc -p -s AppInfo
4092 ShellExperienc SearchUI.exe "C:\Windows\SystemApps\ShellExperienceHost_cw5n1h2txyewy\ShellExperienceHost.exe" -ServerName:AppXtk181tbx0c2q5x02s8tw7hfxa9xb3t.mca
2036 RuntimeBroker C:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
1052 svchost.exe C:\Windows\system32\svchost.exe -k netsvc -p -s WpnService
1704 RuntimeBroker C:\Windows\System32\RuntimeBroker.exe -Embedding
4200 RegSvc.exe "C:\Users\admin\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\update.exe"
1960 svchost.exe C:\Windows\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s PcaSvc
2064 svchost.exe C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s StorSvc
4332 w3wp.exe -
1676 taskhostw.exe taskhostw.exe
900 updatenow.exe "C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe"
3628 FRK Imager.exe C:\Program Files\AccessData\FRK Imager\FRK Imager.exe
1564 svchost.exe C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s WdiSystemHost
4656 wermgr.exe -
4648 taskhostw.exe -
3184 taskhostw.exe -
81875 -
```

Q8 — Reverse Shell Host Process & PID

Q8 The reverse shell outbound traffic is handled by a built-in Windows process that also spawns the implanted executable. What is the process name and PID?

Answer w3wp.exe | PID 4332

Method Command: vol.py -f memdump.mem windows.netscan — Filtered for port 4443 / remote IP 10.0.2.4. The IIS worker process w3wp.exe (PID 4332) was found managing the established outbound connection, confirming it as both the reverse shell host and parent process for updatenow.exe.

```
kn999@kn999: ~/volatility3
Session Actions Edit View Help
kn999@kn999: ~/volatility3
tem32\svchost.exe -k netsvcs -p -s IKEEXT C:\Windows\system32\svchost.exe
** 1680 628 svchost.exe 0xce0656eb4080 4 - 0 False 2024-09-10 05:29:34.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s FontCache C:\Windows\system32\svchost.exe
** 1808 628 svchost.exe 0xce0656f94080 5 - 0 False 2024-09-10 05:29:38.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k NetworkService -p -s LanmanWorkstation C:\Windows\System32\svchost.exe
** 3856 628 svchost.exe 0xce0657309080 7 - 0 False 2024-09-10 05:32:27.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s CDPSvc C:\Windows\system32\svchost.exe
** 2064 628 svchost.exe 0xce06578ed080 6 - 0 False 2024-09-10 05:44:23.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s StorSvc C:\Windows\System32\svchost.exe
** 660 628 svchost.exe 0xce0656cc30c0 5 - 0 False 2024-09-10 05:29:22.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s NcbService C:\Windows\System32\svchost.exe
** 2452 628 svchost.exe 0xce06571cb280 15 - 0 False 2024-09-10 05:30:04.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s w3wp.exe C:\Windows\system32\svchost.exe
** 4332 2452 w3wp.exe 0xce06574ca080 0 - 0 False 2024-09-10 06:10:48.000000 UTC \Device\HarddiskVolume1\Windows\System
32\inetstrvw3wp.exe -p -s - 0
*** 900 4332 updatenow.exe 0xce0657ddb1c0 3 - 0 True 2024-09-10 06:08:23.000000 UTC N/A \Device\HarddiskVolume1\ProgramData\Microsoft\Windows\Start Me
nu\Programs\Startup\updatenow.exe "C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe"
** 792 628 svchost.exe 0xce0654f69080 14 - 0 False 2024-09-10 05:29:12.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k DcomLaunch -p C:\Windows\system32\svchost.exe
** 2696 792 SearchUI.exe 0xce065730d080 14 - 1 False 2024-09-10 05:33:42.000000 UTC N/A \Device\HarddiskVolume1\Windows\SystemApps\Microsoft.Windows.C
ortana_cw5n1h2txyewy\SearchUI.exe "C:\Windows\SystemApps\Microsoft.Windows.Cortana_cw5n1h2txyewy\SearchUI.exe" -ServerName:CortanaUI.AppXa50dqa5gqv4a428c9y1jjw7a3btvepj.mca C:\Win
dows\SystemApps\Microsoft.Windows.Cortana_cw5n1h2txyewy\SearchUI.exe
** 1784 792 RuntimeBroker.exe 0xce06579ad080 1 - 1 False 2024-09-10 05:34:01.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\RuntimeBroker.exe C
:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
** 4092 792 ShellExperienc 0xce065772b080 13 - 1 False 2024-09-10 05:33:35.000000 UTC N/A \Device\HarddiskVolume1\Windows\SystemApps\ShellExperienceHost
_cw5n1h2txyewy\ShellExperienceHost.exe "C:\Windows\SystemApps\ShellExperienceHost_cw5n1h2txyewy\ShellExperienceHost.exe" -ServerName:App.AppXtk181tbxbce2qsex02s8tw7hfxa9xb3t.mca C:\Win
dows\SystemApps\ShellExperienceHost_cw5n1h2txyewy\ShellExperienceHost.exe
** 2036 792 RuntimeBroker.exe 0xce0656dd40c0 3 - 1 False 2024-09-10 05:33:46.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\RuntimeBroker.exe C
:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
** 2072 628 svchost.exe 0xce065703e080 8 - 0 False 2024-09-10 05:29:52.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s UserManager C:\Windows\System32\svchost.exe
** 2172 2072 sihost.exe 0xce06571d5080 5 - 1 False 2024-09-10 05:31:50.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\sihost.exe sihost
.exe C:\Windows\system32\sihost.exe
** 1308 628 svchost.exe 0xce0656ddc080 3 - 0 False 2024-09-10 05:29:27.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s Themes C:\Windows\System32\svchost.exe
** 412 628 svchost.exe 0xce065791a080 1 - 0 False 2024-09-10 05:33:32.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s AppInfo C:\Windows\system32\svchost.exe
** 1052 628 svchost.exe 0xce0657783080 7 - 0 False 2024-09-10 05:34:00.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s UsbSvc C:\Windows\system32\svchost.exe
** 1564 628 svchost.exe 0xce0657a4f080 2 - 0 False 2024-09-10 06:12:41.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s WdiSystemHost C:\Windows\System32\svchost.exe
** 1440 628 svchost.exe 0xce06562e0080 7 - 0 False 2024-09-10 05:29:29.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalServiceNetworkRestricted -p -s Dhcp C:\Windows\System32\svchost.exe
** 2296 628 svchost.exe 0xce0654d5a080 1 - 0 False 2024-09-10 05:30:04.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k utcsvc -p C:\Windows\System32\svchost.exe
** 1316 628 svchost.exe 0xce0656dda080 7 - 0 False 2024-09-10 05:29:27.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s EventSystem C:\Windows\system32\svchost.exe
** 2340 628 svchost.exe 0xce065711c300 1 - 0 False 2024-09-10 05:30:04.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\sys
```

The malware sample (updatenow.exe) was examined via static analysis (strings/FLOSS) and threat intelligence platforms (VirusTotal, MalwareBazaar). SHA-256 hash was used as the primary pivot.

c25a6673a24d169de1bb399d226c12cdc666e0fa534149fc9fa7896ee61d406f

Q9 Static inspection reveals the binary has been packed to hinder analysis. Which packer was used?

Community Score **-66**

updatesnow.exe

peexe detect-debug-environment checks-network-adapters calls-wmi upx long-sleeps checks-bios

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY (18+)

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Basic properties ⓘ

MDS	d79760029d6dbed4497725579d814b7e
SHA-1	54b22078651dcee13472858d706f86a34678b2e
SHA-256	c25a6673a24d169de1bb399d226c12cdc66e0fa534149fc9a7896ee61d406f
Vhash	06503cf07d70101011z11z673101013z13z10101019z
AuthenticHash	9cef1fcc216ca2d5b819765042ac1f1d1a182583fd45e6799e4c0ec4e1e560cf
Imphash	fc6683d30d9f2544a50fd5357825e79
Rich PE header hash	a5d888b5a108c327d65f490cc1a712f2
SSDEEP	12288-NYV6MorX7qzuC3QH09FQVHPF5ljgcQTimgHdJ0BNajn5iBXu9HGavHNhdUun5
TLSH	T19AD422C11FC1D85AC0A923B9CA3FDD0098067CB2DBE53B1E8658F45AF866783D417AAE
File type	Win32 EXE executable windows win32 pe peexe
Magpie	PE32 executable (GUI) Intel 80386, for MS Windows, UPX compressed
TrID	UPX compressed Win32 Executable (35.7%) Win32 EXE Yoda's Crypter (35%) Win32 Dynamic Link Library (generic) (8.6%) Win16 NE executable (generic) (6.6%) Win32 Executable (generic) ...
DetectItEasy	PE32 Packer: UPX (3.91) [NRV,brute] Library: Autolt (3.XX) Compiler: Microsoft Visual C/C++ (18.00.40629) [POGO_O_CPP] Linker: Microsoft Linker (12.00.40629) Tool: Visual Studio (2013)
Magika	PEBIN
File size	588.00 KB (602112 bytes)
PEID packer	UPX v0.89.6 -v1.02 /v1.05 -v1.24 → Markus & Laszlo [overlay]
Varist packer	UPX

History ⓘ

Q10 — C2 FQDN

Q10 Threat-intel analysis shows the malware beaconing to its C2 host. Which FQDN does it contact?	
Answer	cp8nl.hyperhost.ua
Method	VirusTotal Relations tab → Contacted Domains revealed cp8nl.hyperhost.ua with 6 security vendor detections. Strings/FLOSS analysis also surfaced this Ukrainian hosting domain as an embedded C2 callback address.

cp8nl.hyperhost.ua

2/91

Community Score

2/91 security vendors flagged this domain as malicious

Reanalyze More

cp8nl.hyperhost.ua

hyperhost.ua

Registrar

ua.ukrnames

Last Analysis Date

8 hours ago

DETECTION

DETAILS

RELATIONS

COMMUNITY 3

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Crowdsourced context

HIGH 1

MEDIUM 0

LOW 0

INFO 0

SUCCESS 0

Activity related to AGENTTESLA

 - according to source Cluster25 - 2 years ago
This DOMAIN is used as a CnC by AGENTTESLA. AgentTesla is spyware designed in .Net to steal personal data, capture screenshots, and record clipboard and form values. It's marketed on its own website, falsely claimed to be a legitimate keylogger. Popular in the hacker community for its ease of use and tech support, it likely originated in Turkey.

Security vendors' analysis

Do you want to automate checks?

ADMINUSLabs	Malicious	CyRadar	Malicious
alphaMountain.ai	Spam	Abusix	Clean

Q11 — Malware Family

Q11 Open-source intel associates that hash with a well-known commodity RAT. To which malware family does the sample belong?

Answer AgentTesla

Method	VirusTotal Detection tab (multiple vendors) and MalwareBazaar both attributed the sample to AgentTesla — a .NET-based information-stealer and RAT capable of credential harvesting, keylogging, screenshot capture, and clipboard monitoring. The malware is sold commercially, making it a commodity threat.
---------------	---

🔍 c25a6b73a24d169de1bb399d226c12cdc666e0fa534149cf9fa7896ee61d406f

[Sign in](#)
[Sign up](#)

59

/71

Community Score -66

ⓘ 59/71 security vendors flagged this file as malicious

[Reanalyze](#)
[Similar](#)
[More](#)

c25a6b73a24d169de1bb399d226c12cdc666e0fa534149cf9fa7896ee61d406f
updateNOW.exe

peexe
detect-debug-environment
checks-network-adapters
calls-wmi
uxp
long-sleeps
checks-bios

Size
588.0 KB

Last Analysis Date
2 days ago

EXE

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[COMMUNITY](#)
18+

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Popular threat label ⓘ trojan.autoit/strab

Threat categories trojan downloader

Family labels autoit strab formbook

Security vendors' analysis ⓘ
Do you want to automate checks?

Vendor	Detection	Category	Action
Alibaba	ⓘ Trojan:Win32/Strab.0fbca37d	AliCloud	ⓘ Trojan.Win.AgentTesla.SHZT
ALYac	ⓘ Trojan.GenericKD.73970718	Antiy-AVL	ⓘ Trojan/Win32.Formbooks
Arcabit	ⓘ Trojan.Generic.D468B41E	Arctic Wolf	ⓘ Unsafe
Avast	ⓘ Script:SNH-gen [Trj]	AVG	ⓘ Script:SNH-gen [Trj]
Avira (no cloud)	ⓘ TR/W32.Evo	BitDefender	ⓘ Trojan.GenericKD.73970718

5. MITRE ATT&CK Mapping

Complete mapping of observed attacker behaviors to the MITRE ATT&CK Enterprise framework:

Phase	Technique ID	Technique Name	Observable
Reconnaissance	T1046	Network Service Discovery	Port scan from 10.0.2.4 → 10.0.2.15
Initial Access	T1190	Exploit Public-Facing Application	Web shell upload via SMB share
Execution	T1059.001	Command & Scripting: PowerShell/Cmd	shell.aspx web shell executed on IIS
Persistence	T1547.001	Startup Folder	updatenow.exe in Windows Startup folder
Defense Evasion	T1027.002	Software Packing (UPX)	updatenow.exe packed with UPX
Discovery	T1046	Network Service Discovery	SMB Tree Connect enumeration
Lateral Movement	T1021.002	SMB/Windows Admin Shares	SMB share access: \\10.0.2.15\Documents
C2	T1571	Non-Standard Port	Reverse shell on port 4443
C2	T1071	Application Layer Protocol	AgentTesla beaconing to cp8nl.hyperhost.ua
Collection	T1056.001	Input Capture: Keylogging	AgentTesla keylogger capability
Exfiltration	T1041	Exfil Over C2 Channel	Credential/data theft via AgentTesla C2

6. Indicators of Compromise (IoCs)

Type	Indicator	Description
IP Address	10.0.2.4	Attacker source IP — reconnaissance & C2
IP Address	10.0.2.15	Victim IIS server
File	shell.aspx	Web shell uploaded via SMB (1,015,024 bytes)
File	updatenow.exe	Packed AgentTesla RAT — persistence implant
File Path	C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe	Startup folder persistence path
SHA-256	c25a6673a24d169de1bb399d226c12cdc666e0fa534149fc9fa7896ee61d406f	updatenow.exe hash
Domain (C2)	cp8n1.hyperhost.ua	AgentTesla C2 callback domain
Port	4443/TCP	Reverse shell listening port
Process	w3wp.exe (PID 4332)	IIS worker process hosting reverse shell
UNC Path	\\10.0.2.15\Documents	Enumerated SMB share
UNC Path	\\10.0.2.15\IPC\$	Enumerated SMB IPC share
Packer	UPX	Packer used for defense evasion
Malware Family	AgentTesla	Commodity RAT — credential/keylogger

7. Attack Chain Narrative

Step 1 — Reconnaissance: The attacker at 10.0.2.4 began with aggressive port scanning against the IIS server (10.0.2.15), sending rapid-fire SYN probes to identify open services (T1046).

Step 2 — Initial Access via SMB: After discovering an accessible SMB service, the attacker enumerated shares (\\10.0.2.15\Documents, \\10.0.2.15\IPC\$) and uploaded shell.aspx (1,015,024 bytes) to a web-accessible directory.

Step 3 — Web Shell Execution: The ASPX web shell was triggered via an HTTP request to the IIS server, providing remote code execution in the context of the IIS worker process (w3wp.exe, PID 4332).

Step 4 — Reverse Shell: The web shell was used to establish a reverse TCP shell back to the attacker on port 4443, a port commonly permitted outbound through perimeter firewalls.

Step 5 — Persistence: The w3wp.exe process spawned updatenow.exe, dropping the payload into the Windows Startup folder (T1547.001) to ensure execution on every system boot.

Step 6 — Defense Evasion & C2: The binary was packed with UPX (T1027.002) to hinder static analysis. Once running, AgentTesla beacons to cp8nl.hyperhost.ua for C2 communication, enabling credential theft, keylogging, and remote control.

8. Recommendations

- Block and isolate 10.0.2.4 at the perimeter firewall; review all outbound connections to port 4443 and the domain cp8nl.hyperhost.ua.
 - Immediately remove shell.aspx from the IIS server and audit all files in web-accessible directories.
 - Remove updatenow.exe from the Startup folder and verify no additional persistence mechanisms exist (registry run keys, scheduled tasks).
 - Restrict SMB access to the IIS server; audit share permissions on \\10.0.2.15\Documents.
 - Reset all credentials on the compromised host; AgentTesla is a credential harvester — assume all stored passwords, cookies, and keys are compromised.
 - Implement endpoint detection for UPX-packed binaries and alert on w3wp.exe spawning non-standard child processes.
 - Enable egress filtering to block non-standard outbound ports (e.g., 4443) unless explicitly required by business functions.
-